Water-to-Energy Campus — Burke Lakefront Airport, Cleveland.

This is a genuinely exciting convergence — one of the most significant urban redevelopment opportunities in the Midwest, perfectly sized and situated for a world-class water-to-energy campus. Let me lay out the full vision.

The Site — What Burke Actually Offers

Burke Lakefront Airport presents a once-in-a-century opportunity to repurpose roughly 450 acres of lakefront land for public access and job-creating development. It's possible to build on the landfill beneath Burke Lakefront Airport, and the land beneath the runways is compact enough to support structures. The site was described as remarkably clean compared with other redeveloped industrial sites. However, a construction crew could dig down only 12 feet before hitting the water table, limiting structures to park space and one- or two-story buildings in some areas.

This shallow water table constraint is not a liability for a water-to-energy campus — it is a direct asset. Lake Erie freshwater is essentially at grade. The electrolyser intake, cooling water supply, and process water systems can draw from the lake with minimal pumping head. No other industrial site in Ohio offers this combination: 450 acres of flat, cleared, serviced land with unlimited freshwater at the boundary and direct connection to downtown Cleveland's power grid and district heating potential.

A city-commissioned study noted that closing Burke and redeveloping it with recreation and limited residential/commercial space could generate up to $92 million per year in new economic activity. A water-to-energy campus would go significantly beyond this figure by adding industrial energy output, green hydrogen export, and campus anchor employment.

Now here is the full master plan. Every zone is clickable. Here are the detailed engineering and economic cases for each.

Why This Site Is Exceptional for Water-to-Energy

The convergence of site characteristics at Burke is essentially unique in the continental United States. Lake Erie provides unlimited soft water freshwater — unlike the Great Salt Lake or coastal sites, Erie water has low TDS (typically 150–200 mg/L), low hardness, and minimal pretreatment requirements before electrolysis. The flat, already-cleared 450-acre footprint cuts the grading, demolition, and remediation costs that burden most industrial sites. The land beneath the runways is compact enough to support structures, and the site was described as remarkably clean compared with other redeveloped industrial sites.

Downtown Cleveland's existing power grid, district steam infrastructure, and the Waterfront Line light rail station at East 9th Street provide immediate connection to energy markets and workforce.

The city and county described Burke as a once-in-a-century opportunity to repurpose roughly 450 acres of lakefront land for public access and job-creating development. A water-to-energy campus transforms this opportunity from passive real estate play into an active energy production asset — one that generates revenue from day one of operation rather than waiting for residential and retail tenants to fill out over decades.

Zone 1 — Lake Erie Water Treatment + PEM Electrolyser Farm (~25 acres)

The water treatment plant is the foundation of everything. Lake Erie water is drawn via a submerged intake structure anchored to the existing breakwater — the Army Corps of Engineers already maintain infrastructure here for the dredge disposal facility, providing a regulatory pathway for a new intake permit.

Treatment training for electrolyser-grade water:

- Traveling screens and debris filters (the lake carries significant suspended solids from storm events)
- Ultrafiltration (UF) membranes — removes biologicals, algae, and fine particulates.
- Two-stage RO — reduces TDS from ~180 mg/L to <10 mg/L
- Mixed-bed DI polishing — achieves <0.1 μS/cm conductivity required by PEM stacks.
- Chemical dosing (antiscalant, chlorination/dichlorination)

At 100 MW of PEM electrolyser capacity, water consumption is approximately 90,000 liters/hr — comparable to a modest municipal water plant. The RO brine reject (~25,000 L/hr) is returned to the lake at a diffuser found at least 200 meters from the intake.

The 100 MW PEM farm produces approximately 2,000 kg H_2/hr (48 tons/day). At $3/kg, this is $144,000/day in hydrogen commodity value before any power generation. Surplus renewable electricity from Ohio's expanding wind and solar portfolio (FirstEnergy and AES Ohio have major Lake Erie wind development plans) feeds the electrolysers at off-peak prices — potentially below $25/MWh — making green hydrogen production economics increasingly competitive.

Zone 2 — FastOx Gasification Campus (~40 acres including waste reception)

The FastOx array processes Cleveland and Northeast Ohio's municipal solid waste — currently landfilled at significant cost to the region — converting it to syngas, then clean hydrogen, with vitrified stone and recovered metals as byproducts.

Northeast Ohio's eight-county region generates approximately 3,000–4,000 tons of MSW per day. A 500–1,000 t/day FastOx installation at Burke captures 12–25% of this stream, diverting it from Subtitle D landfills and converting it to approximately 70–140 MW of hydrogen energy equivalent. The gate fee revenue — typically $30–60/ton in Ohio — adds $15,000–$60,000/day in tipping fee income, making this the lowest-cost hydrogen feedstock of any process on the campus.

The gasification campus includes enclosed waste reception (negative-pressure tipping hall), two to four FastOx shaft gasifiers in parallel, a syngas polisher, waste heat boiler, HCl and sulfur guard beds, HTS/LTS shift reactors, MDEA CO_2 absorber, and methanation. The CO_2 captured from the shift reaction is either piped to the ammonia swing plant for urea production or sold to greenhouse operators and food-grade users in the region.

Vitrified stone output (~100 kg/ton MSW) — 50–100 tons/day — is sold as road aggregate or construction fill, closing the material loop entirely.

Zone 3 — KARNO Power Island + H_2 Turbine CCGT (~30 acres)

This is the generation heart of the campus. The total hydrogen supply — from electrolysis plus FastOx gasification — supports 120–240 MW of hydrogen-equivalent energy per hour. This feeds hybrid generation architecture:

The KARNO array (50–100 MW, 250–500 modules across 25–50 containers) provides continuous baseload generation with near-zero emissions, 4,000-hour no-touch maintenance intervals, and multi-fuel flexibility. The KARNO's flameless oxidation handles the full H_2 output from electrolysis without combustion hardware modifications. Hot-side heat (at 70–90°C) feeds the district heating loop in Zone 5.

The H_2 turbine CCGT (20–40 MW, using a Kawasaki L30A or Siemens SGT-800 class turbine) provides the campus's peaking and grid services capacity — rapid ramp, frequency response, and capacity market participation. The HRSG captures turbine exhaust heat for added district heating contribution and drives a steam turbine for supplemental power.

Net campus power output, after internal parasitic loads:

Source	Gross MW	Parasitic	Net MW
KARNO array	50–100	8–15	42–85
H_2 CCGT turbine	20–40	2–4	18–36
Campus total	70–140	10–19	60–121

At $50/MWh average grid price, 60–121 MW net export generates $72,000–$145,000/day in power revenue alone.

Zone 4 — Hydrogen Buffer and Export Hub

The campus produces hydrogen from two sources (electrolysis and FastOx) at varying rates. A shared buffer storage system — bank of 80-bar pressure vessels sized for 4–8 hours of electrolyser output — decouples production from consumption and export schedules.

Export routes:

- High-pressure tube trailer filling station (200–250 bar) — serves industrial H_2 customers in the Cleveland manufacturing corridor (steel, chemicals, refining)
- Pipeline connection — a 6-inch H_2 pipeline running along the existing East 9th Street corridor connects to the industrial Flats district and potential Cleveland Cliffs steel operations.
- Ammonia swing plant — a small Haber-Bosch module (50–100 t/day NH_3) converts surplus H_2 to ammonia when hydrogen spot prices are depressed, providing a price floor and serving Ohio's significant agricultural fertiliser market.

Ohio is the fifth-largest corn-producing state in the US. A green ammonia production capability at Burke would position Cleveland as a strategic supplier to the Midwest agricultural belt — a market currently served entirely by fossil-fuel-derived ammonia from Gulf Coast facilities.

Zone 5 — District Energy and Grid Substation

The KARNO array rejects approximately equal thermal energy to its electrical output — at 75 MW electrical, this is ~75 MW thermal available as 70–90°C hot water. This is not waste heat in the conventional sense; it is a second revenue stream. A district heating network running under the proposed Shoreway boulevard redesign connects the Burke campus to downtown Cleveland's office towers, the Convention Center, and the lakefront residential zone — replacing gas-fired boilers with zero-emission heat from hydrogen oxidation.

Cleveland's existing district steam system (operated by Enterprise/Clearway in the downtown core) provides a ready-made connection point. The Burke CHP district heating network would effectively extend and decarbonize this existing system.

The grid substation replaces the conventional utility substation model with a KARNO microgrid PCC, as detailed in the earlier conversation. The campus is fully island-capable — in a major grid outage (ice storm, severe weather on Lake Erie, substation fault), the Burke campus continues generating power and can be operated as a grid-forming resource to support critical downtown Cleveland loads.

Zone 6 — Public Waterfront Park (150+ acres)

Burke Lakefront Airport is an obstacle. You can't build a decent-sized park on the lakefront because of it. You can't build lakefront high-rises because of it. And you can't access much of the water's edge near Downtown Cleveland because of it.

The water-to-energy campus occupies only the western ~250 acres of the site (Zones 1–5). The eastern ~150+ acres become a genuine public waterfront park — something Cleveland has sought for generations. The park includes a continuous lakefront promenade connecting North Coast Harbor to the east, a kayak and paddleboard launch, a public marina with slips for Lake Erie sailing, an event lawn for the National Air Show (which can continue without the airport, as Chicago demonstrated after closing Meigs Field), and a centerpiece Energy Education Pavilion.

The pavilion is the visitor-facing element of the energy campus — a public attraction that makes the hydrogen production and KARNO generation infrastructure visible and legible to the public. Transparent panels showing electrolysers in operation, interpretive exhibits on water-to-energy conversion, and an observation deck overlooking the KARNO container array and Lake Erie. This is both civic infrastructure and the most compelling clean energy tourism attraction between Chicago and New York.

Zone 7 — Innovation District and Residential (~100 acres)

The city and county described closing Burke as creating a rare, contiguous stretch of lakefront land for redevelopment, parks, housing, and public space.

The southern portion of the campus (landward of the Shoreway, which is redesigned as a boulevard with intersections rather than a limited-access highway) hosts the innovation district. The 12-foot depth-to-water-table constraint limits building heights to one or two stories in portions of the site, but this actually suits the desired campus character — low-density, green-roof buildings housing clean energy R&D firms, Case Western Reserve and Cleveland State University research partnerships, and the National Renewable Energy Laboratory's potential Midwest hub.

Key anchor tenants:

- A 10–20 MW KARNO-powered data center — one of the first genuinely zero-emission data centers in North America, cooled by Lake Erie water, powered by on-site green hydrogen generation. This alone is a nationally significant asset at a moment when data center power demand is the defining infrastructure challenge of the decade.
- A green hydrogen and fuel cell workforce training center — directly addressing Ohio's manufacturing workforce transition, building on Cleveland's existing manufacturing heritage.
- Mixed-income residential ground-floor retail, built on the compact-fill portions of the site.

Overall Campus Economics

Revenue stream	Daily	Annual
Power export (60–121 MW @ $50/MWh)	$72K–$145K	$26M–$53M
H_2 export (surplus @ $3/kg)	$50K–$100K	$18M–$37M
Gate fees (500–1,000 t/day MSW @ $45/t)	$22K–$45K	$8M–$16M
District heat sales	$15K–$30K	$5M–$11M
NH_3 / urea sales	$10K–$25K	$4M–$9M
Data centre lease revenue	$20K	$7M
Park / marina / pavilion	$5K–$10K	$2M–$4M

Revenue stream	Daily	Annual
Total gross revenue	**\$194K–\$375K**	**\$70M–\$137M**

This compares favorably with the city's commissioned study projection of $92M/year from mixed-use residential and retail development alone — and the energy campus revenues begin flowing from day one of plant commissioning, years before a residential development would reach full occupancy.

Regulatory and Phasing Pathway

Phase 0 (now–2027): FAA decommissioning process proceeds in parallel. Zone 2 FastOx and Zone 5 substation can begin permitting as industrial facilities on the eastern lakebed lease portion even before full closure — using existing port authority industrial land designations.

Phase 1 (2027–2029): Airport closes. Zone 1 water treatment and first 500-tonne FastOx unit commissioned. First 50 KARNO modules (10 MW) operational. District heating pilot to North Coast Harbor.

Phase 2 (2029–2031): Full 100 MW PEM electrolyser farm operational. KARNO array expanded to 50 MW. H_2 turbine CCGT commissioned. Public Park opens. Energy Education Pavilion opens.

Phase 3 (2031–2034): Innovation district and data center operational. Full 100 MW KARNO array. NH_3 swing plant. Full district heating network to downtown core. Campus reaches design revenue.

A city-commissioned study found that closing Burke and developing the land it sits on would be the most economically productive use of that space. A water-to-energy campus makes that economic case dramatically stronger — it is not just a real estate development but a permanent, revenue-generating energy infrastructure asset that positions Cleveland as a global leader in green hydrogen and clean power at the precise moment when those industries are entering their most rapid growth phase.

Would you like to continue with **detailed site civil and foundation engineering for the landfill substrate**, **the Lake Erie water intake and environmental permitting**, **the grid interconnection and Cleveland utility coordination**, or **a full project finance and investment structure**?